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Palaeomagnetism and chronology of the central Taupo Volcanic Zone, New Zealand

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SUMMARY

The central Taupo Volcanic Zone (TVZ) of New Zealand is a region of intense Quaternary silicic volcanism, active since 1.6 Ma. We report palaeomagnetic measurements from 59 distinct volcanic units sampled at 98 sites in the TVZ. These are mainly rhyolitic ignimbrites and lava domes, with a few basaltic, andesitic, and dacitic lavas. Most have new K/Ar or $^{40}Ar/^{39}Ar$ ages. The remanent magnetizations are generally stable to both thermal and alternating-field demagnetization, and well-determined mean palaeodirections were obtained for all sites.

Our findings suggest that the Taupo, Whakamaru, Maroa, Reporoa, Rotorua, and Okataina volcanic centres were magnetized during the Brunhes normal chron. Kapenga is an older volcanic centre, where activity commenced around 0.89 Ma and extended into the Brunhes. Mangakino volcanic centre is significantly older and was active from 1.6 to 0.95 Ma.

Transitional or intermediate palaeodirections were obtained from Ahuroa ignimbrite (1.18 ± 0.02 Ma) and Mamaku ignimbrite (0.22 ± 0.01 Ma). The former almost certainly corresponds to the Cobb Mountain Event. The latter is significantly older than the Blake Event, and probably corresponds to the recently reported Pringle Falls/Summer Lake magnetic episode.

Multiple sites from the Whakamaru ignimbrite have indistinguishable $^{40}Ar/^{39}Ar$ ages (0.33 ± 0.01 Ma) and glass composition, but divergent palaeomagnetic directions. This contrast suggests that either (1) the different sites were formed during a phase of extremely violent activity, lasting up to a few hundred years, during which geomagnetic secular variation was recorded; or (2) that they were formed in a single eruption, and rotation during subsequent extensional tectonism has caused divergence of the palaeodirections.

$^{40}Ar/^{39}Ar$ ages of 0.77 ± 0.03 Ma for the reversely magnetized Rahopeka ignimbrite and 0.71 ± 0.06 Ma for the overlying normally magnetized Waiotapu ignimbrite bracket and constrain the age of the Matuyama–Brunhes transition.

Key words: geochronology, igneous rock, palaeomagnetism, Taupo Volcanic Zone, volcanic activity.

INTRODUCTION

The Taupo Volcanic Zone (TVZ) covers a region 125 km long and 60 km wide in the central North Island of New Zealand. The central part of the TVZ is one of the most productive silicic volcanic systems on Earth and has erupted more than 10^4 km^3 of magma over the past 1.6 Ma (Houghton *et al.* 1995). By contrast, the northern and southern parts of the TVZ contain the active, andestic volcanoes of White Island and Tongariro. At least 34 major rhyolitic–ignimbrite eruptions are thought to have formed an overlapping succession of eight

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caldera or volcanic centres within the central TVZ (Fig. 1). In addition, there have been a number of phases of lava-dome formation, and minor basaltic, andesitic and dacitic lavas are also found (Wilson *et al.* 1995). The identification and eruptive histories of the various volcanic centres are described in several volcanostratigraphic studies (Cole 1979; Wilson *et al.* 1984; Wilson, Houghton & Lloyd 1986; Nairn, Wood & Bailey 1994). The Rotorua, Okataina, Reporoa, Maroa and Taupo volcanic centres have all formed since 250 ka, and portions of the caldera margins are recognizable modern topographic features. The Whakamaru, Kapenga and Mangakino volcanic centres are older structures, largely obscured by younger faulting, volcanism and sedimentary fill, and have been identified from geophysical and volcanological evidence. This palaeomagnetic study was conducted in parallel with detailed new K/Ar and $^{40}Ar/^{39}Ar$ geochronology (Houghton *et al.* 1991; Pringle *et al.* 1992; Houghton *et al.* 1995): in many instances palaeomagnetic and dating samples were taken from the same sites. The $^{40}Ar/^{39}Ar$ data yield an age of 1.55 ± 0.05 Ma for ignimbrite A, the oldest known TVZ ignimbrite, some 500 ka older than previous estimates based on fission-track methods. Early palaeomagnetic studies of TVZ volcanic units include a pioneering study by Hatherton (1954) on the magnetic properties of the Whakamaru ignimbrite, and more broadly based studies of the remanent magnetization, susceptibility and angular dispersion produced by geomagnetic secular variation by Cox (1969, 1971). Cox's study included 14 units from the central TVZ, and a further 17 units from the southern TVZ and Northland, Coromandel and Taranaki. A number of other studies conducted over the past two decades have also included palaeomagnetic measurements (e.g. Murphy & Seward 1981; Soengkono *et al.* 1992).

Cox (1969, 1971) found only normal polarities from the TVZ units that he studied, and deduced that volcanic activity post-dated the Matuyama–Brunhes transition, then dated at 0.68 Ma. However, Cox sampled only a few of the numerous different units now distinguished. Furthermore, many of his samples were not demagnetized in the detailed manner customary nowadays, so it is unlikely that he always isolated the primary direction of magnetization, particularly if it was of reversed polarity. Murphy & Seward (1981) reported reversed polarities from a number of previously unrecognized ignimbrites exposed in the Matahana Basin, and obtained a fission track age of 1.03 Ma for the oldest, the Tikorangi ignimbrite. A large number of reversely magnetized units have now been discovered in the west of the region (Tanaka, Houghton & Turner 1991; this study); with the new isotopic ages now available, these confirm significant activity in the central TVZ prior to the Matuyama–Brunhes reversal. Soengkono *et al.* (1992) have tried to determine the extent of reversely magnetized rocks in the western central TVZ in order to model aeromagnetic data.

The purpose of the present study is to extend the initial work of Cox to as many of the known ignimbrites and other key TVZ units as possible. By integrating the new palaeomagnetic and geochronological results, we are able to constrain further the histories of each of the volcanic centres, and of the TVZ as a whole. We use our data to confirm the correlations of certain units, suspected on geological and chemical grounds to represent the same eruption. In other cases, however, correlative units yield divergent palaeomagnetic directions, and these are interpreted either in terms of geomagnetic palaeosecular variation or in terms of the extensional tectonism of the region.

PROCEDURES

Samples were taken at 98 sites in 59 cooling units older than 0.05 Ma. At most sites eight independently oriented samples were drilled. 11 of the units sampled duplicate those studied by Cox (1969). The units sampled comprise 33 ignimbrites (32 welded and one non-welded pyroclastic flow), 20 rhyolite lava domes, and six basaltic or andesitic lavas. Three were from Taupo volcanic centre, 12 from Whakamaru, 10 from Maroa, 16 from Mangakino, four from Kapenga, eight from Rotorua, two from Reporoa, and four from Okataina (Fig. 1a). The only major units that were not sampled were four non-welded ignimbrites associated with phreatomagmatic eruptions from the Taupo and Mangakino centres. The sampling sites are designated by the prefix 'NT' and are shown in Fig. 1(a), together with the inferred locations of the eight main volcanic centres. Fig. 1(b) is a simplified geological map redrawn from Cole (1979), in which ignimbrites originating from the TVZ are distinguished from others and from the Mesozoic basement. Unit names, ages, latitude and longitude and NZ grid references (1 : 50 000 series) of the sites are summarized in Appendix A, together with the palaeomagnetic results.

Palaeomagnetic measurements were made at both Tokyo Institute of Technology and Victoria University of Wellington. In most cases, two pilot specimens from each site were demagnetized at each laboratory: one by stepwise alternating field (AF) and one by stepwise thermal demagnetization (i.e. at least four specimens from each site were demagnetized in detail). At most sites three to four more specimens were AF demagnetized to an appropriate peak field, usually between 10 and 25 mT, to remove secondary components of magnetization. However, at sites which showed intermediate palaeodirections or non-convergent natural remanent magnetization (NRM) directions, all specimens were carefully AF demagnetized in a progressive manner.

PALAEOMAGNETIC RESULTS

Secondary components, probably viscous remanent magnetization (VRM) acquired in the present-day field, are small at most sites and were usually removed in the first few steps of demagnetization. All samples show a high degree of stability to both AF and thermal demagnetization. Typical median destructive fields (MDFs) lie between 20 and 35 mT, while blocking temperatures (T_b) generally range from 300 ° to 580 °C, but with the blocking temperature spectra of a few specimens extending to about 640 °C. Examples of orthogonal component plots (Zijderveld 1967) for AF and thermal demagnetization are shown in Fig. 2. The resistance to demagnetization and the high stability of NRM to both AF and thermal demagnetization, illustrated in Fig. 2(a) and (b) for normal (Whakamaru, site NT05) and reversed (Marshall, site NT54) units, are consistent with the assumption that the samples carry a thermoremanent magnetization (TRM). Fig. 2 also includes examples of within-site sample directions. As seen in Fig. 2(c), which includes directions from two sites (NT05 and NT54) on the same projection, after partial demagnetization to remove secondary components, sample directions are usually tightly grouped, giving typical alpha-95 values (semi-angle of cone of

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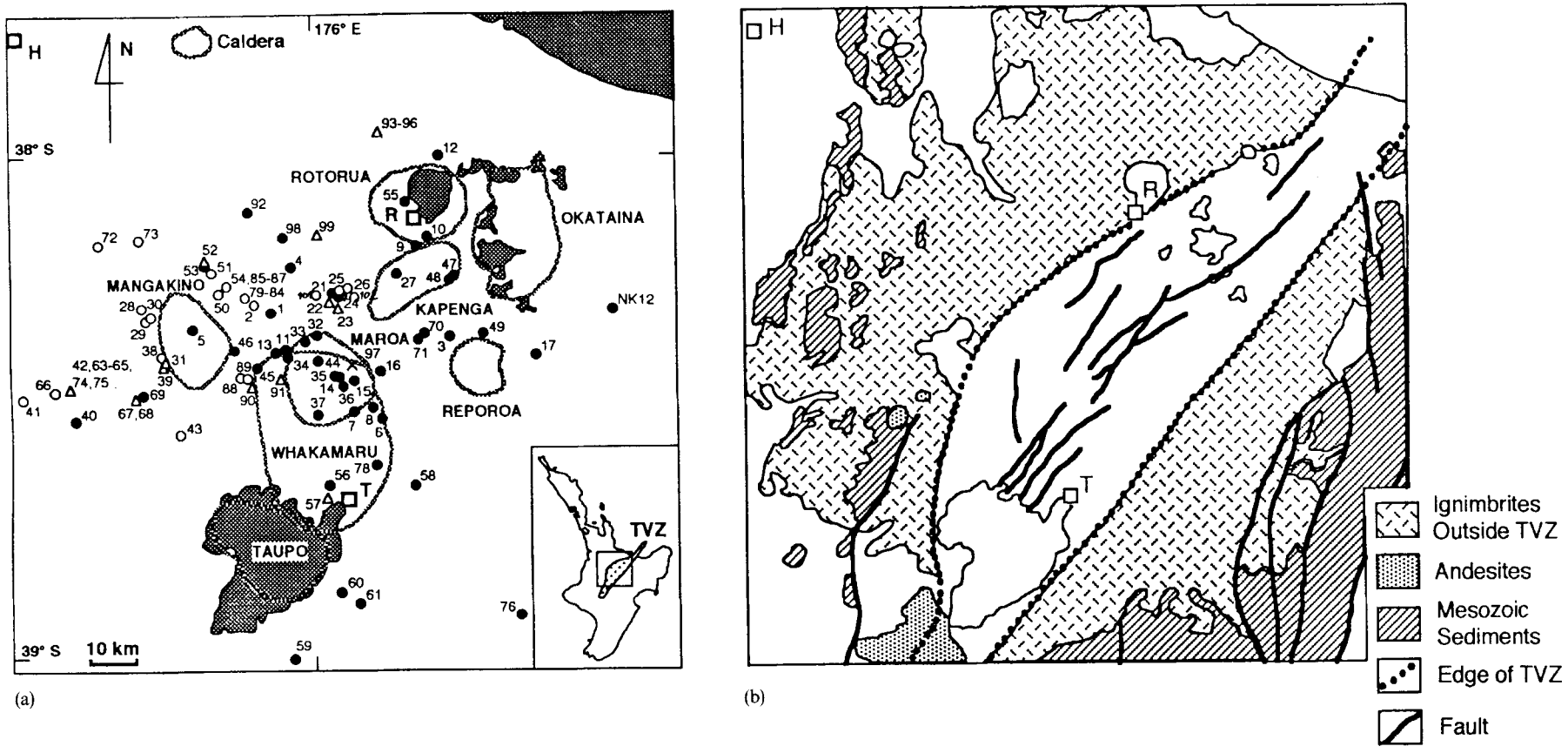


Figure 1. (a) Site locality map of the Taupo Volcanic Zone. Bold lines show the inferred outlines of the volcanic centres (after Houghton *et al.* 1987). Solid circles, open circles, and open triangles show sites where normal, reversed, and transitional primary directions were obtained, respectively. The cross indicates an unsuccessful site. Open squares show major cities: H — Hamilton, R — Rotorua, T — Taupo. Shaded areas indicate major lakes and the Bay of Plenty coastline. (b) Simplified geology of the Taupo Volcanic Zone, including major fault systems.

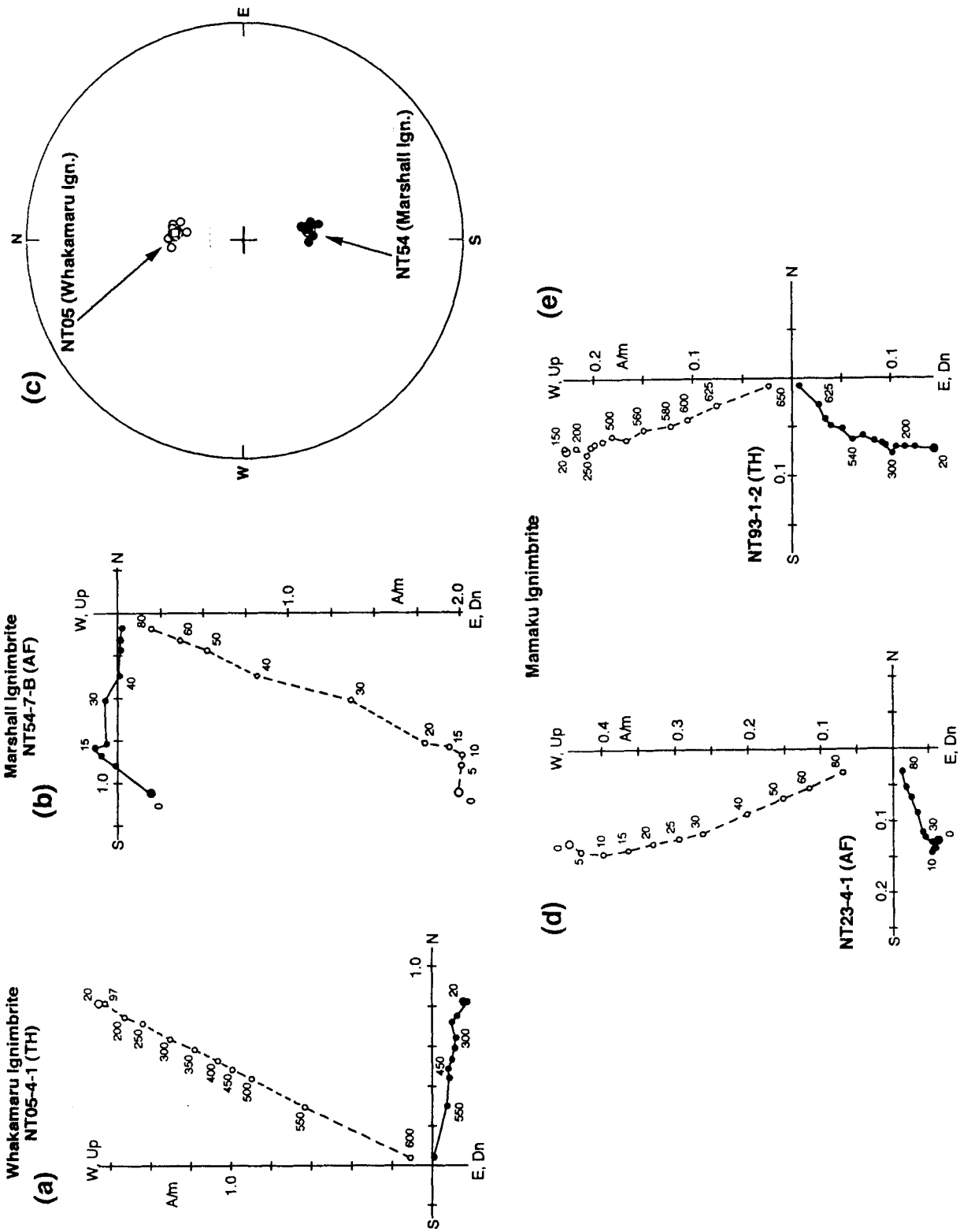


Figure 2. Examples of orthogonal component plots of alternating field (AF) and thermal (TH) demagnetization results and equal-area stereographic plots of remanence directions for samples with normal and reversed polarity (a, b, c) and transitional (d, e, f, g, h) primary directions. Solid and open circles on the orthogonal component plots indicate projections on the horizontal and vertical planes, respectively. On the equal-area plots, solid and open circles indicate downward and upward directions respectively. (i) and (j) show an example in which a large secondary component hinders the isolation of a primary direction on the orthogonal plot, but on the equal-area projection successive remanent directions show a good fit to a great circle. (k) illustrates the combined analysis of four endpoint directions and two great-circle fits obtained from samples from this site.

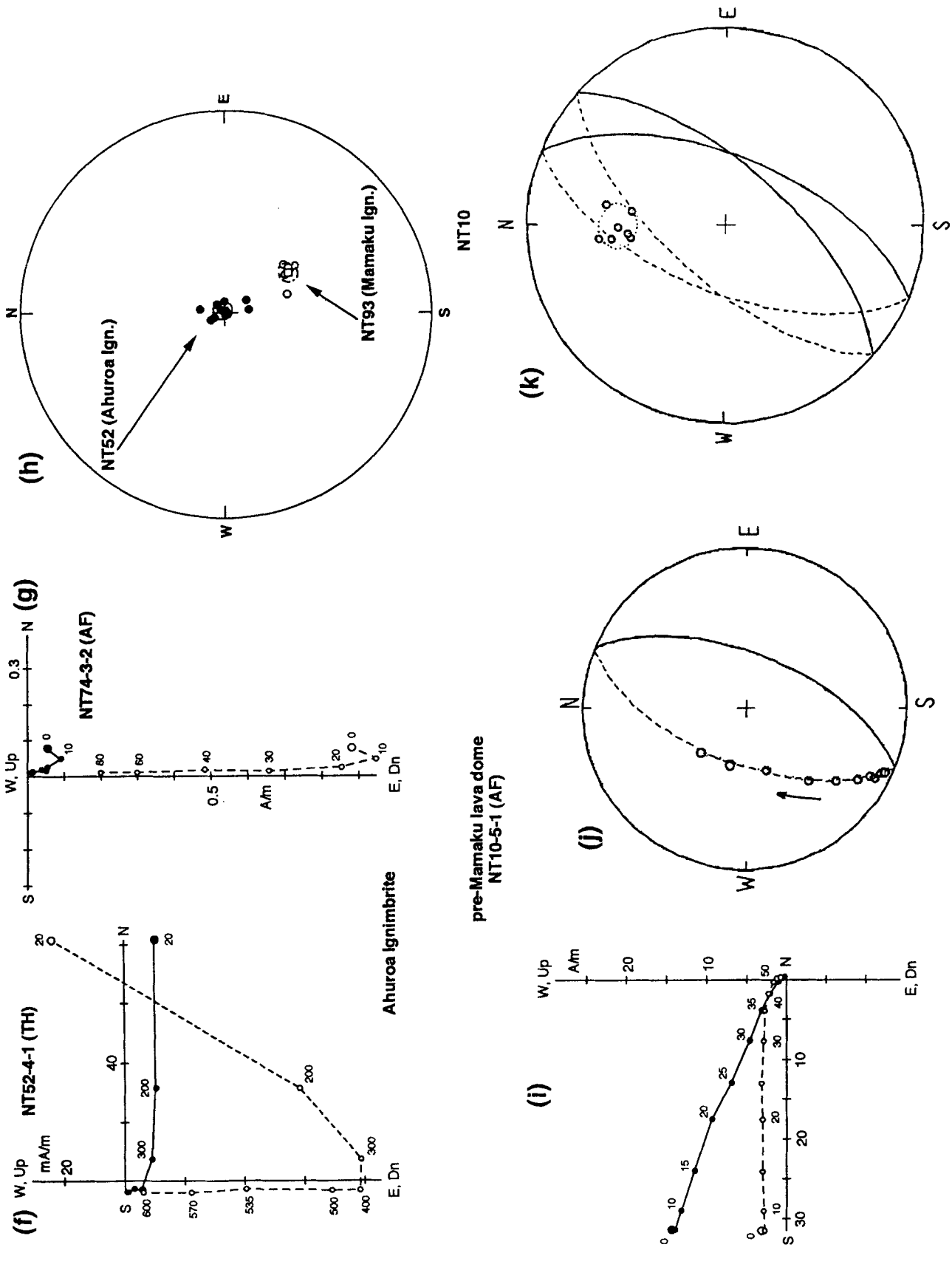


Figure 2. (*Continued.*)

95 per cent confidence in mean direction) between 1° and 8°. Most of the major units yielded palaeomagnetic directions which were clearly of normal or reversed polarity. Figs 2(d) to (h), however, show examples of plots from two important ignimbrites, Mamaku (sites NT23 and 93) and Ahuroa (sites NT52 and 74), from which intermediate directions (i.e. latitude of the virtual geomagnetic pole $<45°$) were obtained.

At most sites the blocking temperature and coercivity spectra are consistent with titanomagnetite as the carrier of the primary remanence. Fig. 2(e), however, shows a specimen from the Mamaku ignimbrite that is typical of a few specimens in which the blocking-temperature spectrum extends beyond 600 °C. In these specimens, a component of the remanent magnetization is carried by a mineral with a Curie Point higher than that of titanomagnetite or magnetite (580 °C). Rock magnetic work in progress (Turner, private communication) indicates that this is due to a spinel-structured, cation-deficient titanomagnetite, which is thermally stable. This might have been formed by late deuteric oxidation of primary titanomagnetite during initial cooling or during subsequent weathering processes. Similar compositions have been suggested by Schlinger, Veblen & Rosenbaum (1991), Rosenbaum (1993) and McIntosh (1991) to explain observations in tuffs and ignimbrites from Nevada and New Mexico. Rosenbaum (1993) also notes that variations in the amount and grain-size distribution of microcrystals of the high-blocking-temperature mineral result in pronounced variations in magnetic properties with vertical position in some flows. Magnetic inhomogeneity was studied in the Whakamaru ignimbrite by Hatherton (1954), and is considered likely throughout the TVZ ignimbrites, highlighting the necessity for careful examination of progressive demagnetization results.

In the case of these Mamaku specimens, the difference between the directions of the high- and mid-blocking-temperature components is very small. More significantly, consistent intermediate directions were obtained from all six sites at which the Mamaku ignimbrite was sampled. An example from site NT93 is shown in Fig. 2(h). The average Mamaku palaeomagnetic direction obtained from eight sites is $D=149.4°$, $I=-68.3°$, $\alpha\text{-}95=6.2°$, corresponding to a Virtual Geomagnetic Pole (VGP) position in the equatorial Atlantic.

The intermediate directions obtained from the Ahuroa ignimbrite are shown in Figs 2(f) to (h). A large viscous component in the direction of the present-day field was observed at some sites (e.g. NT52-4-1; Fig. 2f), while other sites from the same unit contain almost no secondary component (Fig. 2g). Sample directions are well grouped within each site, as typified by site NT52, shown in Fig. 2(h). The mean direction obtained from the 10 sites at which the Ahuroa ignimbrite was sampled is $D=10.4°$, $I=80.4°$, $\alpha\text{-}95=7.5°$, yielding a VGP to the north of New Zealand.

Fig. 2(i) shows the result of AF demagnetization of a sample from a lava dome near the rim of the Rotorua volcanic centre, which pre-dates the Mamaku eruption (site NT10). In this case, a persistent secondary component of magnetization made it difficult to isolate a primary endpoint. Successive remanence directions, however, fall on a well-defined great-circle path (Fig. 2j), which was determined using the method of principal component analysis, as described by Kirschvink (1980). At this site, great-circle paths were calculated for two specimens and, using the method of McFadden & McElhinny (1988), these were combined with stable endpoint directions from four further specimens to calculate a site mean and α-95 (Fig. 2k).

Unit mean palaeodirections and VGPs for all sites are shown in Figs 3(a) and (b), respectively. The mean palaeomagnetic directions and corresponding VGPs calculated from between six and 10 samples at each site are tabulated in Appendix A. Of the 98 sites or levels sampled, 47 yielded palaeomagnetic directions of normal polarity, 30 yielded reversed directions, nine, intermediate directions with VGPs in the Atlantic Ocean, and 12, intermediate directions with VGPs in the Pacific, to the NE of New Zealand. The relevance of the intermediate directions is further discussed below. The mean of the normal directions is $D=7.9°$, $I=-57.9°$, $\alpha\text{-}95=6.5°$. The mean reversed direction is $D=178.5°$, $I=62.4°$, $\alpha\text{-}95=7.1°$. The reversed mean is indistinguishable from the direction expected of a reversed axial geocentric dipole (AGD) ($D=180°$, $I=57.7°$), but the normal mean is slightly to the east of the normal AGD direction ($D=0°$, $I=-57.7°$). The reason for this small difference is uncertain: the time spanned by the data set should be long enough to average out palaeosecular variation. Furthermore, any overall tectonic rotation would be expected to affect the older, reversed data set at least as much as the younger, normal one.

MAGNETOSTRATIGRAPHY AND REVISED CHRONOLOGY

Using the new palaeomagnetic data and chronology, it is possible to construct a magnetostratigraphic history for the eight volcanic centres of the central TVZ (Fig. 4). The polarity assigned to each unit is based on VGP latitude (P_{lat}): normal for $P_{lat}>45°N$; reversed for $P_{lat}>45°S$; otherwise intermediate. The majority of units are normally magnetized. The notable exceptions are most of the ignimbrites from Mangakino and the older units from the Kapenga volcanic centre, which are reversed, and the Ahuroa and Mamaku ignimbrites, Whakaahu lava dome (south) and K-Trig basalt, all of which record intermediate directions.

We deduce that Mangakino was active exclusively during the later part of the reversed Matuyama chron, that Kapenga became active during the last part of the Matuyama and activity continued well into the normal Brunhes chron, while the other centres did not begin activity until sometime after the Matuyama–Brunhes reversal.

Our site-mean directions and those of Cox (1969) usually agree to within about 10°. In most cases, the difference is not significant at the 95 per cent level of confidence; in some, the difference may be due, at least in part, to incomplete removal of secondary magnetizations by Cox (1969). While he checked for the presence of secondary components using stepwise AF demagnetization, most of the site means quoted by Cox (1969) were calculated by averaging NRM directions. In particular, Cox would not have found the transitional direction of the Ahuroa ignimbrite because he did not demagnetize specimens from this unit. We note also that Cox's unit NZ-12 was incorrectly named as Marshall ignimbrite in his study. The unit has since been correlated with the Waiotapu ignimbrite (0.71 Ma), which is normally magnetized, while the Marshall ignimbrite is reversed. Discrepancies in magnetization direction between different sites of a given unit do, however, sometimes arise for natural reasons. These include differential post-emplacement rotation, due to active tectonism, plastic deformation below the blocking temperatures of the magnetic minerals, and cooling over prolonged time periods, during

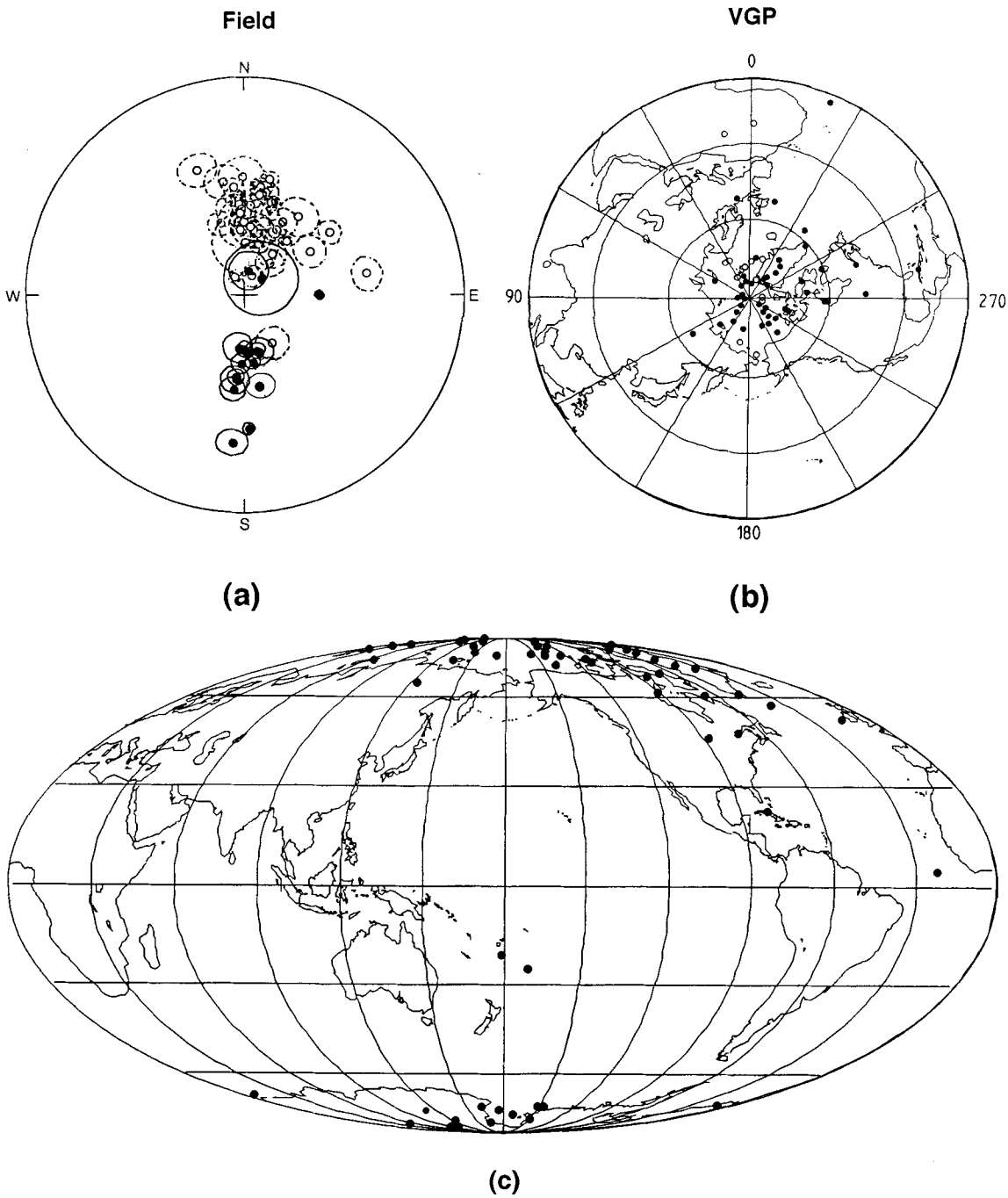


Figure 3. (a) Equal-area projection of unit mean palaeodirections and cones of 95 per cent confidence (α-95s) from all 59 units sampled. Open symbols = upper hemisphere, closed symbols = lower hemisphere; (b) corresponding virtual geomagnetic poles (VGPs) plotted on a polar equal-area projection of the northern hemisphere; closed symbols = north poles (normal polarity), open symbols = south poles (reversed polarity). (c) Virtual geomagnetic (north) poles plotted on a (Mollweide) projection of the whole world.

which the magnetic field direction has changed due to palaeosecular variation. These effects are further discussed below.

The paragraphs below give a brief description of each of the TVZ volcanic centres and discuss the relevance of our new data to their formation and eruptive history.

Mangakino volcanic centre

Surface evidence for the presence of the Mangakino volcanic centre is largely obscured by younger overlying ignimbrites, which originated further to the east. However, its shape is defined by a major negative gravity anomaly (Rogan 1982) caused by the low density of the fill. Mangakino is inferred as the source of seven major welded and two non-welded ignimbrites (Table 1), six of which are found over wide areas of the King Country to the west. Prior to the recent dating and palaeomagnetic studies, fission-track ages and field constraints indicated that the volcano was active from 1.1 to about 0.5 Ma. (Wilson *et al.* 1984), making it the oldest identified caldera volcano of the TVZ [Wilson *et al.* (1984) discounted an earlier

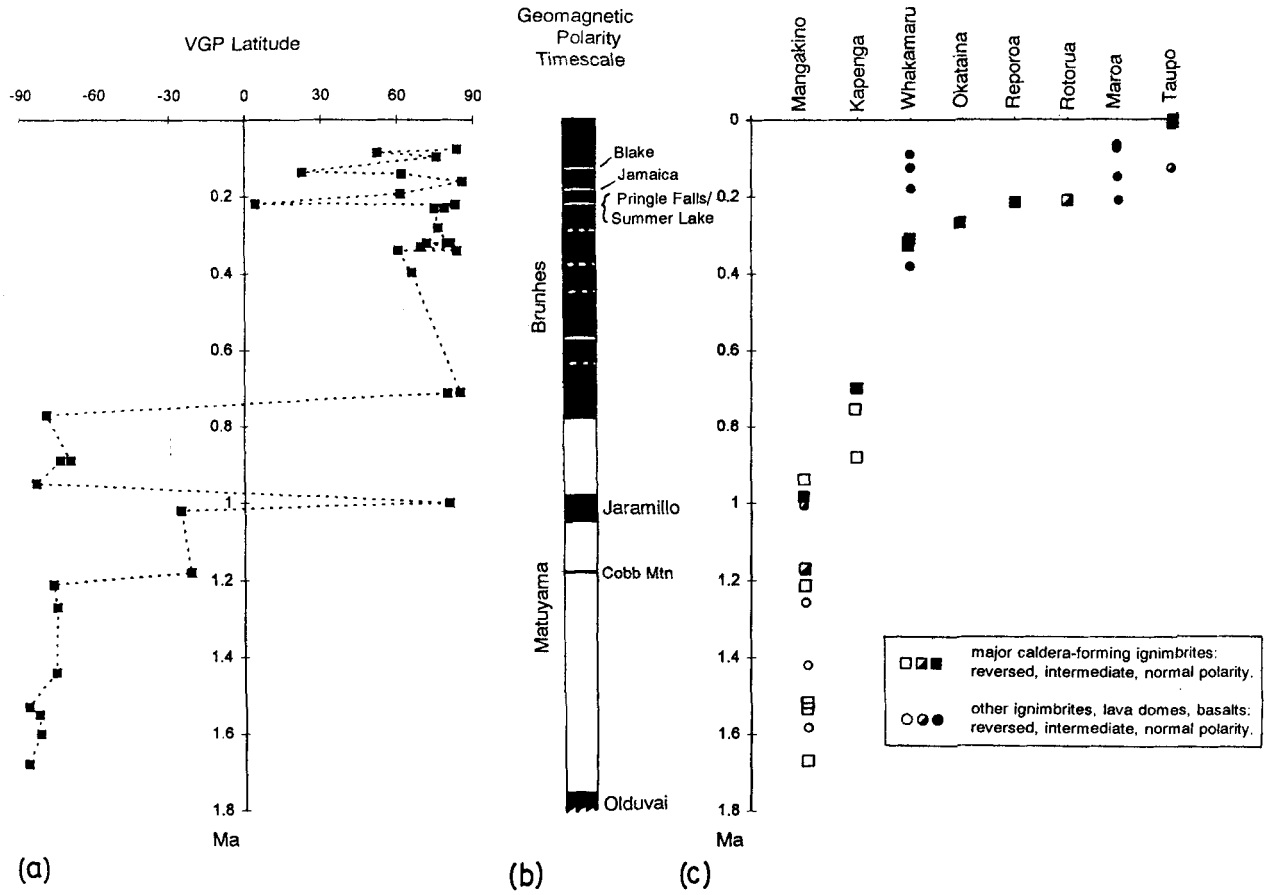


Figure 4. Summary diagram of the magnetostratigraphy of the Taupo Volcanic Zone. (a) Latitude of the virtual geomagnetic pole plotted against age for all dated units. (b) Geomagnetic polarity timescale, from Cande & Kent (1992), with positions of the events proposed by Champion *et al.* (1988), together with the Pringle Falls/Summer Lake event (Hererro-Bervera *et al.* 1994; Negrini *et al.* 1994) and the Cobb Mountain Event. (c) Units arranged by associated volcanic centre, horizontally from left to right, according to age of formation. Closed, open, and half-closed symbols indicate normal, reversed, and transitional palaeomagnetic directions, respectively. Major caldera-forming ignimbrites are shown as squares, other volcanic units as circles. (Rolles Peak andesite is not shown on this figure, since it is not associated with one of the major calderas. Units without age control are not included.) The two recent eruptions from the Taupo centre (*c.* 20000 BP and 1850 BP), although not sampled in this study, have been added for completeness.

Table 1. Ignimbrites originating from Mangakino volcanic centre.

Unit	Previous Age	Previous Polarity	New Age (Ma)	New Polarity
Marshall Ig.	$0.52 \pm 0.14^{M\&S}$	N^{C}	0.95 ± 0.03	R
Rocky Hill Ig.	0.31 ± 0.06^{K}	N	1.00 ± 0.05	N
Ahuroa Ig.	0.65 ± 0.09^{K}	N	1.18 ± 0.02	I
Waipari Ig. *				R
Ongatiti Ig.	0.75 ± 0.08^{K}	R	1.21 ± 0.04	R
Wharepuhunga Ig.			1.44 ± 0.03	R
Ignimbrite C			1.67 ± 0.12	R
Ignimbrite B			1.53 ± 0.04	R
Ignimbrite A	1.10^{R}	R	1.55 ± 0.05	R

* suspected to represent the closing phase of the Ongatiti eruption.
[C] Cox, 1969
[K] Fission Track ages from Kohn, 1973
[M&S] Fission Track ages from Murphy and Seward, 1981
[R] Fission Track age from Rutherford, 1976

fission-track age of 0.31 Ma (Kohn 1973) for the Rocky Hill Ignimbrite as it was out of sequence]. The new $^{40}Ar/^{39}Ar$ ages reported by Houghton *et al.* (1995) show that the Mangakino volcanic centre is even older, active from 1.6 to 0.95 Ma. Our new palaeomagnetic results show that all Mangakino-derived units are of Matuyama age, in accord with this new chronology.

The Marshall ignimbrite is dated at 0.95 ± 0.03 Ma and is reversely magnetized, while the stratigraphically lower Rocky

Hill ignimbrite, dated at 1.00 ± 0.04 Ma, is normally magnetized. The Rocky Hill ignimbrite was therefore erupted during the Jaramillo normal subchron (1.05–0.98 Ma; Cande & Kent 1992). The 1.18 ± 0.02 Ma Ahuroa ignimbrite records an intermediate direction and is stratigraphically below the Rocky Hill ignimbrite. Because of the 0.18 Ma age difference between the Rocky Hill and Ahuroa ignimbrites, it is unlikely that the Ahuroa eruption could have occurred during the lower Jaramillo transition. We believe that the intermediate directions in the Ahuroa correspond to the Cobb Mountain subchron. Turrin, Donnelly-Nolan & Hearn (1994) have recently obtained an $^{40}Ar/^{39}Ar$ age of 1.186 ± 0.006 Ma for the Cobb Mountain subchron, recorded at Alder Creek, California. Our date on the Ahuroa ignimbrite is in excellent agreement with this age, and is significantly older than the previously accepted K–Ar-based age of 1.12 ± 0.02 Ma (Mankinen, Donnelly & Gromme 1978; Mankinen & Gromme 1982). The new ages also support the astronomically tuned revision of the polarity timescale (Shackleton, Berger & Peltier 1990; Hilgen 1991a, 1991b). The older Ongatiti ignimbrite and ignimbrites C, B and A are all reversely magnetized and their ages are consistent with eruption during the reversed interval of the Matuyama chron, between the Olduvai and Jaramillo subchrons.

Our new data assist in correlating some of the older, poorly exposed units at Mangakino. The palaeomagnetic directions obtained from the confirmed Mangakino units and some of their suspected correlatives are shown on an equal-angle projection in Fig. 5(a). Wilson *et al.* (1986) suggest, on chemical and mineralogical grounds, that the Waipari ignimbrite may represent the closing stages of the Ongatiti eruption. The Ongatiti ignimbrite is widespread and voluminous, and the eruption was complex: the 6.5° difference between the palaeomagnetic directions recorded at the two Ongatiti sites is not significant, and the Waipari direction falls between them. Thus, there is no palaeomagnetic reason to suggest that the Ongatiti and Waipari eruptions were separated in time. The older ignimbrites A, B and C form a group with lower inclinations, ignimbrites B and C having indistinguishable directions, and ignimbrite A having a very well-defined, slightly steeper direction. The Wharepuhunga ignimbrite has been suspected to be correlative with ignimbrite B or C, but this is unlikely as it

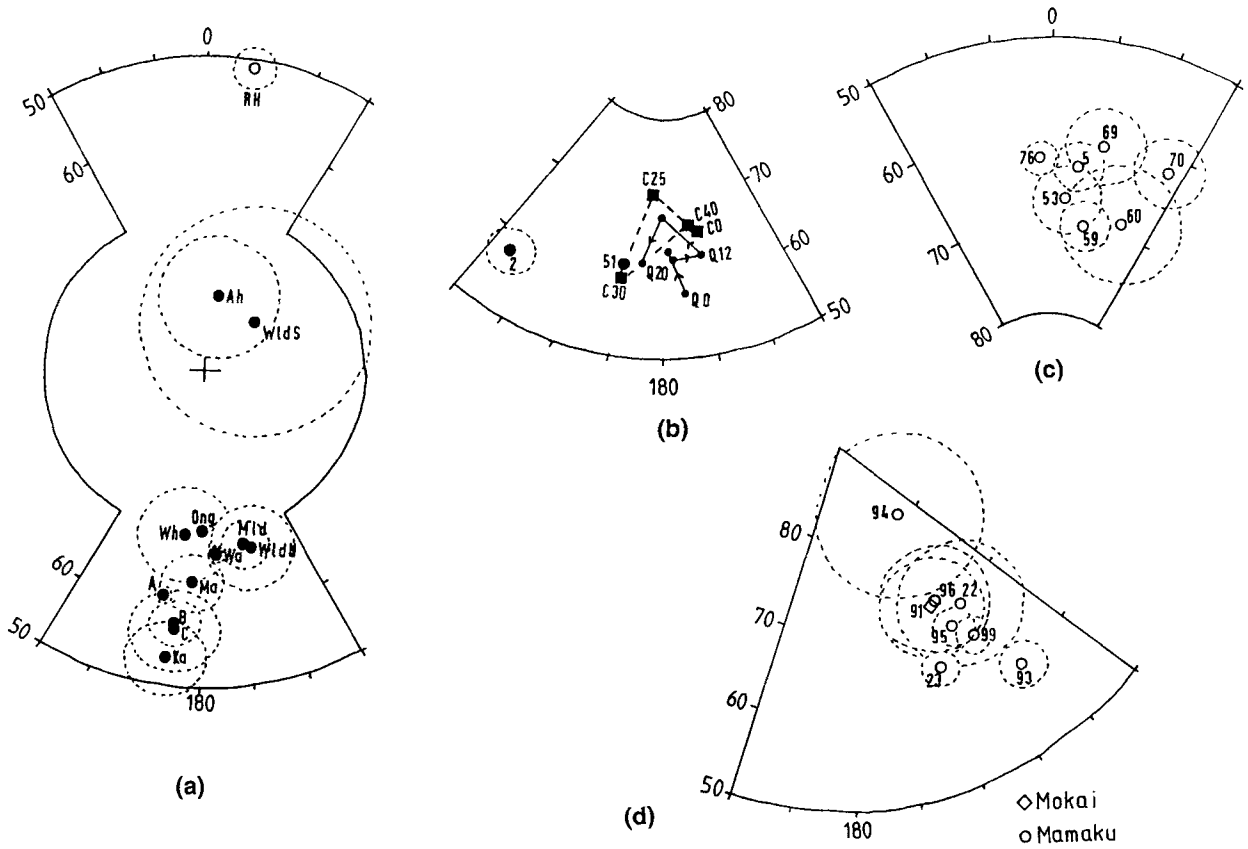


Figure 5. Equal-area projections. (a) Unit mean palaeodirections and cones of 95 per cent confidence (α-95) for ignimbrites and lava domes of Mangakino volcanic centre: Ah = Ahuroa ignimbrite (mean of 10 sites), Wld (S), (N) = Whakaahu lava dome, south (NT90), north (NT89), Ong = Ongatiti ignimbrite (mean of NT41 and 66), Wa = Waipari ignimbrite (NT73), A, B, C = ignimbrites A (mean of NT28 and 30), B (NT29) and C (NT38), Wh = Wharepuhunga ignimbrite (NT72), Ma = Marshall ignimbrite (mean of 12 sites/levels), Mld = Mangakino lava dome (NT50), RH = Rocky Hill ignimbrite (NT40). (b) Mean palaeodirections from different levels within the Marshall ignimbrite: Q = six levels sampled at the Quarry Road locality (NT79-84), and elevation in flow; C = four levels sampled at the Collie Road locality, and elevation in flow. NT02 and NT51 are two further sites at which the Marshall ignimbrite was sampled. For clarity, α-95s are not shown; they are listed in Appendix A. (c) Site mean palaeodirections and α-95s for Whakamaru Group ignimbrites: (NT)05 = Whakamaru ignimbrite, (NT)76 = Te Whaiiti ignimbrite, (NT)53 and (NT)69 = Mananui ignimbrite, (NT)59 and (NT)60 = Rangitaiki ignimbrite, (NT)70 = Paeroa ignimbrite. (d) Site/level mean palaeodirections and α-95s for the Mamaku and Mokai ignimbrites: NT93-96 = four levels within Mamaku ignimbrite sampled at Oha locality; NT22, 23, 99 = other Mamaku sites as shown in Fig. 1; NT91 = Mokai ignimbrite.

has a much steeper inclination, plotting closer to the Ongatiti and Waipari directions.

Prior to this study, no lava domes had been assigned to the Mangakino centre, but the Whakaahu Dome Belt lies close to its inferred eastern margin. The two Whakaahu lava-dome sites (NT 89 and NT 90) yield distinctly different palaeomagnetic directions. The southern site (NT 90) records an intermediate direction very close to that of the Ahuroa ignimbrite, with a VGP to the NE of New Zealand, while the northern site (NT 89) is reversely magnetized. These results, and the K–Ar age of 1.02 ± 0.01 Ma from the southern site, show that they are much older than the adjacent Western Dome Belt, which is associated with the Whakamaru caldera, and that they are related to the activity of Mangakino volcanic centre. The difference between their palaeomagnetic directions shows that the two domes are not contemporaneous. The age of 1.02 ± 0.01 Ma for the intermediate magnetization at the southern site (NT 90) rules out a correlation with the Ahuroa ignimbrite and suggests that it may correspond to the Lower Jaramillo transition, the age of which is estimated as 1.049 Ma (Cande & Kent 1992). The close proximity of the Ahuroa and southern Whakaahu VGPs may be a reflection of the similarity between the VGP paths for the Cobb Mountain and Lower Jaramillo transitions. The northern, reversely magnetized dome is almost certainly of Matuyama age, but there is no field evidence to suggest whether it is older or younger than the southern dome, and hence pre- or post-Jaramillo. A recently mapped and partly buried lava dome further north (NT50) has a reversed magnetization indistinguishable from that of NT89, and is dated at 1.27 ± 0.05 Ma. It is also now linked with the activity of the Mangakino centre.

The Marshall ignimbrites were sampled in some detail at two sections in order to determine the variation in palaeomagnetic directions that might be expected within a unit. Houghton, Wilson & Stern (1987) recognized the Marshall ignimbrites, in drillcores from the Tokoroa/Kinleith area, as a compositionally zoned sequence of flow units and deduced that they represent a single eruption, with only minor time breaks between a lower zone, Marshall B, and an upper zone, Marshall A. The correlatives of these two zones have been sampled in this study. At one location (Quarry Road), a 20 m thick section was sampled at six levels (NT 79–84), and at another (Collie Road) a 40 m thick section was sampled at four levels (NT 54, 85–87). Even if erupted over a short period of time, such a thick ignimbrite might be expected to take 10–20 years to cool to ambient temperature if a thermal diffusivity of 6×10^{-7} m^2 s^{-1} is assumed (Jaeger 1968). Accordingly, small directional differences due to geomagnetic secular variation might be recorded between different levels in the flow, or even between different parts of the blocking-temperature spectrum in individual specimens. The palaeomagnetic directions obtained from the Marshall ignimbrite are shown in Fig. 5(b). With the exception of site NT02, they all fall within a cone of semi-angle approximately 5°. Within this cone, there is a suggestion that a repeatable secular variation signal may be recorded: similar directional sequences describing a quasi-anticlockwise loop in Fig. 5(b) are recorded at the two locations. The sequence is recorded in the A zone of the flow at the Quarry Road locality, but in the lower B zone at the Collie Road locality. The shape of the loops is only just defined at the 95 per cent level of confidence. However, if the correlation is significant, it indicates a slower rate of cooling at the thicker Collie Road section. In a detailed study of palaeomagnetic directions through two thick ignimbrites in Nevada, Rosenbaum (1986) reported directional anomalies of up to 50° in the central portion of one flow. He attributed this to plastic deformation below the blocking temperatures of the magnetic minerals. In the second flow, where the effect was not seen, the magnetic minerals had predominantly lower blocking temperatures. A similar explanation is considered unlikely here, since (1) the blocking-temperature spectra are not biased towards high values, (2) the sections are much thinner than those studied by Rosenbaum—20 and 40 m, respectively, as compared with 80 m—and (3) similar variations are observed in two separate sections of the same ignimbrite. The anomalous direction at site NT02 is probably caused by tectonic disturbance at this location.

Kapenga volcanic centre

Like Mangakino, Kapenga is an older volcanic centre obscured by more recent deposits but detected and delineated from geophysical measurements (Rogan 1982; Wilson *et al.* 1984). Houghton *et al.* (1995) suggest that it is a composite feature formed by the superposition of two, or possibly three, caldera volcanoes. Though not positively confirmed, the distribution of the Waiotapu ignimbrite is consistent with a southern source within the centre (Wilson *et al.* 1984). This area is also likely to be the source of two older ignimbrites recognized by Murphy & Seward (1981) in the Matahana Basin: the Pukerimu–Tikorangi and Rahopeka ignimbrites (Wilson *et al.* 1984). There is no weathering break between what Murphy and Seward describe as the Tikorangi and Pukerimu ignimbrites. They are also very similar mineralogically, and their palaeomagnetic directions are indistinguishable, but much shallower than the direction from the Rahopeka ignimbrite. We conclude that they probably represent the same eruption. Following a long quiescent period, activity in the Kapenga area seems to have resumed with two medium-sized ignimbrite eruptions at around 0.2 Ma. Comparisons of previously published and new ages of the major ignimbrites from the Kapenga volcanic centre are summarized in Table 2.

Several recent publications based on astronomical tuning and $^{40}Ar/^{39}Ar$ dating (Shackleton *et al.* 1990; Hilgen 1991a,b; Cande & Kent 1992) have suggested that the Matuyama–Brunhes transition is older than 0.73 Ma, the K/Ar-derived figure that had been generally accepted for the previous two decades (Mankinen & Dalrymple 1979; Berggren *et al.* 1985; Harland *et al.* 1982). Consensus seems to favour a revised date of about 0.78 Ma. At sites NT105 and NT24 we sampled two ignimbrites, one overlying the other, which record reversed and normal polarities, respectively. NT105 is the site at which Murphy & Seward originally sampled the Rahopeka ignimbrite, and which has a new $^{40}Ar/^{39}Ar$ age of 0.77 ± 0.03 Ma. Because of the similarity of its palaeodirection with other nearby sites of the Waiotapu ignimbrite (NT01, NT21), and its similar mineralogy, site NT24 is assumed to be Waiotapu ignimbrite, now dated at 0.71 ± 0.06 Ma. These units therefore bracket the Matuyama–Brunhes transition. Although our dates effectively permit both the old and the revised ages of the transition, the date of 0.77 ± 0.03 Ma on NT105 places a limit on any further revision.

Table 2. Major ignimbrites originating from Kapenga volcanic centre.

Unit	Previous Age (Fission Track)	Previous Polarity	New Age (Ma) ($^{40}Ar/^{39}Ar$)	New Polarity
Waiotapu Ig.		N[C]	0.71 ± 0.06	N
Rahopeka Ig.	0.72 ± 0.16[M&S]	R[M&S]	0.77 ± 0.03	R
Tikorangi Ig.	1.03 ± 0.18[M&S]	R[M&S]	0.89 ± 0.04	R
Pukerimu Ig.	1.03 ± 0.18[M&S]	R[M&S]		R

[C] Cox, 1969
[M&S] Murphy and Seward, 1981

Whakamaru, Maroa and Taupo volcanic centres

Wilson *et al.* (1986) divide the volcanic activity in the region around and to the north of Lake Taupo into three stages of caldera development. The eruption of the voluminous Whakamaru group of welded ignimbrites and the inferred formation of a major caldera centred to the north of the present Lake Taupo occurred at about 0.33 Ma. Subsequent volcanism along the western margin of this centre formed the Western Dome Belt (Houghton *et al.* 1991). Maroa and Taupo are two younger centres, which partially obscure the Whakamaru caldera. Maroa is a caldera associated with the eruption of at least four moderate-volume ignimbrites (Wilson *et al.* 1984, 1986; Rogan 1982) and with a younger, well-developed central-dome complex. The Taupo centre to the south has been vigorously active since 25 ka, with two caldera-forming and 27 smaller eruptions (Wilson 1993). Units from these recent eruptions were not sampled in this study.

The Whakamaru eruption(s) were possibly the largest Late Quaternary eruptive episode in the southern hemisphere. Tephra in deep-sea cores taken up to 10 000 km away have been tentatively associated with the Whakamaru eruption (Froggatt *et al.* 1986). Ignimbrites of the Whakamaru group crop out in two broad groups on the east and west of the TVZ but in the centre of the zone they are downfaulted and buried. The Mananui and Whakamaru ignimbrites found in the west, and the Te Whaiti and Rangitaiki ignimbrites in the east, have similar glass chemistries, and so are likely to be related to the same magma batch. Field evidence, however, suggests that they may have originated in two distinct eruptions (Wilson *et al.* 1986). The Paeroa ignimbrites, found NE of Maroa have also been suggested as Whakamaru lateral equivalents.

The results of this and previous palaeomagnetic studies on the Whakamaru-group ignimbrites are summarized in Table 3 and Fig. 5(c). Although there is overlap between many of the cones of 95 per cent confidence, there is no overwhelming coincidence in the mean palaeomagnetic directions from the different sites (Fig. 5c). Two units from the eastern TVZ, Rangitaiki (sites NT59 and 60) and Paeroa (site NT70) record directions that are steeper and/or to the east of directions from western sites. Even NT69 and NT53, both sites of Mananui ignimbrite, agree only very poorly at the 95 per cent level of confidence. Two explanations are considered for this divergence in the palaeodirections: geomagnetic secular variation and tectonic rotation. Over the last 1000 years the mean rate of change of declination in New Zealand has been about 0.04° a^{-1} (Turner & Lillis 1994). Typical rates of secular variation vary worldwide between 0.01 and 0.1° a^{-1}. In order for palaeo-secular variation to have produced the observed divergence in palaeomagnetic directions, the Whakamaru-group ignimbrites would have had to have been erupted over a period of at least several hundred, possibly several thousand years, and not in a single catastrophic eruption. An alternative explanation is that the palaeodirections were parallel at the time of formation, but have since been rotated by tectonic movement. As noted above, there is some measure of agreement between the palaeo-directions of sites from the same side of the zone. The greatest divergence is between sites on the eastern and western margins respectively: this could have been caused by differential rotation during tectonic extension of the TVZ as a whole. As a simple test of this hypothesis, we looked at the divergence between palaeodirections recorded at different sites of the older Waiotapu ignimbrite (0.71 ± 0.06 Ma). A similar east–west difference is observed with the circular standard deviation from five sites being 7.6°, whereas for the 0.33 Ma Whakamaru ignimbrites it is 5.5° from seven sites. These results are at least consistent with a tectonic model incorporating progressive rotation with time. Further studies are in progress.

Rotorua volcanic centre

The Rotorua volcanic centre is defined by a 20 km diameter basin which formed by collapse, following the eruption of the voluminous Mamaku ignimbrite. The basin is centred on Lake

Table 3. Whakamaru-group ignimbrites.

Unit	Previous Palaeomag. Inc.	Dec.	α-95	New Palaeomag. Inc	Dec	α-95	Site	New age (Ma) ($^{40}Ar/^{39}Ar$)
Mananui Ig.	-65.2	0.6	3.1[C]	-67.8	4.2	3.6	NT53	0.32 ± 0.03
				-61.5	11.5	4.1	NT69	0.32 ± 0.03
Whakamaru Ig.	N[H]			-64.1	6.3	2.6	NT05	0.32 ± 0.02
Te Whaiti Ig.	-65.9	357.4	1.9[C]	-63.4	357.2	1.9	NT46	0.344 ± 0.003
Rangitaiki Ig.	-64.1	357.7	2.7[C]	-70.3	10.2	2.7	NT59	
				-69.1	20.8	6.3	NT60	
Paeroa Ig.				-61.9	26.5	3.7	NT70	0.33 ± 0.01

[H] Hatherton, 1954
[C] Cox, 1969

Rotorua. The only evidence of significant activity before or after the Mamaku eruption is a number of lava domes of uncertain age within and around the caldera margin. Palaeomagnetic results from four sites of Mamaku ignimbrite, including one at which four levels spanning a vertical thickness of 100 m were sampled, yield a consistent, well-defined intermediate direction with an equatorial VGP in the mid-Atlantic region (Fig. 5d). An earlier fission-track age of 0.14 ± 0.08 Ma for the Mamaku ignimbrite (Murphy & Seward 1981) suggested correlation of this intermediate direction with the Blake event, most recently estimated at 0.128 Ma (Champion, Lanphere & Kuntz 1988). Furthermore, the equatorial, Atlantic VGP lies close to many of the published VGP paths for the Blake event (e.g. Tric *et al.* 1991). However, the new $^{40}Ar/^{39}Ar$ age of 0.22 ± 0.01 Ma for Mamaku ignimbrite makes it significantly older than the Blake event. This age is based on analyses of samples from two sites (three if the Mokai ignimbrite is included; see below), and details are given in Table 4. Mamaku is closer in age to the suggested Jamaica/Biwa I episode, which lies between 160 and 200 ka (Ryan 1972; Kawai 1984; Champion *et al.* 1988), but complete, detailed records of this episode do not exist. Recent publications of sedimentary records from Pringle Falls and Summer Lake, Oregon (Hererro-Bervera *et al.* 1994; Negrini *et al.* 1994), however, report a geomagnetic event with a well-defined, repeatable VGP path which heads south from the North Pole, through the Atlantic Ocean and the Americas, before turning north again through the Pacific Ocean, very similar to published VGP paths for the Blake event. The age of this new event is constrained by $^{40}Ar/^{39}Ar$ dates from tephra within the sequence, averaging 218 ± 10 ka. It is therefore thought most likely that the equatorial-Atlantic VGP of the Mamaku ignimbrite corresponds to part of this 'Pringle Falls/Summer Lake' geomagnetic excursion, which may or may not be the same as the previously reported Jamaica/Biwa I event.

On the basis of its appearance and the area it covers, the Mokai ignimbrite (site NT91) has long been suspected to be equivalent to the Mamaku ignimbrite. The intermediate palaeodirection of NT91, which is indistinguishable from the Mamaku direction (Fig. 5d) clearly establishes that the two units are contemporaneous, a conclusion supported by a new $^{40}Ar/^{39}Ar$ age determination of 0.21 ± 0.01 for the Mokai ignimbrite.

Okataina volcanic centre

Adjacent to the Rotorua volcanic centre and active since the eruption of the Matahina ignimbrite at 0.28 ± 0.01 Ma, the Okataina volcanic centre also contains two major dome complexes, Haroharo and Tarawera. Until recently, Okataina was thought to be the source of the Kaingaroa ignimbrites, but these are now recognized as originating from a separate volcanic centre, Reporoa.

Reporoa volcanic centre

Until recently, the Reporoa depression was considered to be a tectonic fault-angle depression. However, the recognition of lithic-lag breccias, interbedded with the Kaingaroa ignimbrites and thinning outwards from the depression, point to Reporoa as the source of the 0.23 ± 0.01 Ma ignimbrites (Nairn *et al.* 1994). Two units of Kaingaroa ignimbrite, I and II, are recognized. These were sampled at sites NT49 and NT17, and have distinct primary directions of magnetization. A third site, NT16, was also correlated tentatively with Kaingaroa II, on mineralogical and geological grounds. However, its extreme easterly declination and steep inclination preclude such a correlation. The identity and source of NT16 remain uncertain at present.

INTERMEDIATE AND ANOMALOUS VGPs

When the VGPs from all sites (Figs 3b and c) are examined carefully, it is apparent that, in addition to the Mamaku and Mokai poles at equatorial latitudes, a number of other VGPs plot in western Europe, the Atlantic Ocean and eastern America, at latitudes lower than normally found during periods of stable polarity. In fact, with the exception of the older Ahuroa ignimbrite and Whakaahu lava dome south, already discussed and attributed to the Cobb Mountain event and lower Jaramillo transition, all VGPs with latitudes below 60° fall between the longitudes of 273° and 8°. Including Mamaku and Mokai, the units with poles falling in this longitudinal band comprise three ignimbrites, three lava domes and two basalts, and they originate from four, or possibly five, of the eight TVZ volcanic centres (Table 5).

Though only the Mamaku, Mokai and K-Trig basalt VGPs can be defined as truly intermediate (VGP latitude $< 45°$), the other five are near the extremes normally associated with secular variation. The observation that they lie in a longitudinally constrained band that includes the Mamaku direction is also possibly significant. The ages of the Ongaroto and K-Trig basalts are not inconsistent with the Blake event, though Mamaku, as noted above, is older. Recent suggestions (e.g. Laj *et al.* 1991, 1992, 1993) that similar transitional field geometries, and therefore similar VGP paths, may persist through several successive reversals are consistent with these units having been erupted during successive periods of abnormal/intermediate field directions.

Recent publications have noted and discussed longitudinally constrained VGP paths (see Jacobs 1994). Tric *et al.* (1991) superimposed VGP paths for the last few reversals and identified preferred bands of paths through the Americas and through eastern Asia and Australia. This has led to renewed suggestions of a dominant dipole component during reversals (Laj *et al.* 1991; Clement 1991), and to claims that the appearance of the bands may be an artefact of the way in which the data are selected and presented (Valet *et al.* 1992;

Table 4. Details of $^{40}Ar/^{39}Ar$ age estimates on Mamaku and Mokai ignimbrites.

Unit	Site	Plateau Age	Isochron Age	$^{40}Ar/^{36}Ar$	MSWD
Mamaku	MP27	0.216 ± 0.008	0.216 ± 0.011	314 ± 5	1.09
Mamaku	MP25	0.224 ± 0.006	0.224 ± 0.013	295 ± 3	1.07
Mokai	MP263	0.212 ± 0.006	0.213 ± 0.022	309 ± 6	0.29

Table 5. Units with VGP latitude less than 60°, excepting Ahuroa and the Whakaahu lava dome.

site	age	volcanic centre	VGP lat	VGP long	δp	δm
Mamaku Ignimbrite(mean)	0.22 ± 0.01^{A}	Rotorua	4.2	337.9	3.1	16.7
Mokai Ignimbrite NT91	0.22 ± 0.01^{A}	Rotorua	5.4	340.6	2.8	16.1
K-trig Basalt NT57	0.137 ± 0.004^{K}	Taupo	22.7	280.1	4.4	7.6
Pre-Mamaku dome NT12		Rotorua	47.5	288.3	7.9	8.1
Ngongotaha dome NT55		Rotorua	45.9	272.6	5.6	4.7
Ongaroto Basalt NT34	0.086 ± 0.02^{K}	Maroa	52.3	346.1	6.0	18.7
Haparangi dome NT27		Kapenga	51.8	7.7	3.6	11.1
NT16		???	57.1	321.6	8.1	10.4

^{A} = $^{40}Ar/^{39}Ar$ age
^{K} = K/Ar age

McFadden 1992). It has also been suggested that a similar banding of VGPs may be present in stable-polarity secular-variation data due to persistent non-zonal terms in the geomagnetic field (Constable 1992). There is some evidence in the TVZ data to support this idea. Normal polarity poles are concentrated between longitudes 270° and 0°. This is most marked for VGPs with latitudes below 70°, and for VGPs above 80° latitude the longitudinal distribution is almost uniform. For the reversed poles, there is a clustering around 180° longitude, the only obvious exceptions being the Pukerimu and Tikorangi poles, which have longitudes of 8.3° and 349.0°, respectively. These data are also consistent with the idea of stationary secular-variation sources, producing a series of repeating waveforms, as observed by Evans (1984), Evans, Gillen & Hedlin (1989) and Negrini *et al.* (1994).

CONCLUSION

Palaeomagnetic data and $^{40}Ar/^{39}Ar$ ages from 59 volcanic units in the Taupo Volcanic Zone have strengthened models for the volcanic history of the region. The Taupo, Whakamaru, Reporoa, Maroa, Rotorua and Okataina volcanic centres were all magnetized during the Brunhes normal polarity chron. Activity at the Kapenga volcanic centre spanned the Matuyama–Brunhes transition, and the Mangakino volcanic centre was active during the Matuyama chron only. The transitional palaeomagnetic direction found in the 1.18 Ma Ahuroa ignimbrite corresponds to the Cobb Mountain subchron and is a rare record from the Southern Hemisphere. The Whakaahu lava dome (S) records a transitional direction that probably corresponds to the Lower Jaramillo reversal. The 0.22 Ma Mamaku/Mokai ignimbrites record an intermediate direction, which is thought to correspond to the recently published Pringle Falls/Summer Lake Excursion. Several other units, including the K-Trig basalt, record low-latitude VGPs in the region of the Atlantic Ocean and the Americas. These may correspond to the same event or the Blake event (or possibly to a continuing magnetic disturbance initiated by one of these events).

The observed divergence between the palaeodirections recorded by the Whakamaru group of ignimbrites may be caused by geomagnetic secular variation recorded over an extended period of eruption or by tectonic rotation associated with post-eruption extension. Further work is needed to distinguish clearly between these interpretations. Due to the likelihood of sites in different parts of the central TVZ experiencing different rotations during the tectonic extension of the region, palaeomagnetic directions are probably only useful in correlating over short distances and/or clearly anomalous directions, for example the Mokai and Mamaku ignimbrites; NT24 and nearby exposures of the Waiotapu ignimbrite.

This paper gives a general overview of an extensive palaeomagnetic study of the central Taupo Volcanic Zone. Because of the breadth of topics covered, it has not been possible to detail each palaeomagnetic, tectonic and rock-magnetic investigation. These will be the subjects of future publications.

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APPENDIX A: SUMMARY OF LOCATIONS, PALAEOMAGNETIC RESULTS (MEAN DIRECTION, ALPHA-95 AND VGP POSITION AT EACH SITE) AND AGE CONTROL FOR SITES IN THE CENTRAL TAUPO VOLCANIC ZONE

Site		Stratigraphic unit	Age (Ma)	Method#	Grid Ref	N	Inc	Dec	α-95	Plat	Plon
Taupo Centre											
NT57	MP53	K-Trig Basalt	0.137+0.004	KA[5]	U18/735763	7	-42.5	79.8	6.1	22.7	280.1
NT56		Punatekahi Basalt			U18/738785	7	-44.2	12.8	3.2	73.3	220.0
NT58	MP60	Rolles Peak Andesite	0.712+0.027	KA[5]	U18/922780	5	-48.5	354.3	3.8	79.7	147.3
Rotorua Centre											
	MP27	Mamaku Ignimbrite	0.22+0.01	AA[1]	U16/786205						
NT22	MP25	Mamaku Ignimbrite	0.22+0.01	c	U16/762192	7	-68.8	145.5	6.8	5.4	335.6
NT23		Mamaku Ignimbrite	0.22+0.01	c	U16/777177	8	-63.6	158.5	2.0	-4.2	341.1
NT93		Mamaku (Oha:100m)	0.22+0.01	c	U15/881559	6	-59.3	142.9	2.6	-4.8	328.6
NT94		Mamaku (Oha: 50m)	0.22+0.01	c	U15/883557	6	-80.3	143.3	7.0	22.2	344.1
NT96		Mamaku (Oha: 20m)	0.22+0.01	c	U15/885555	6	-71.4	151.1	6.0	7.2	340.4
NT95		Mamaku (Oha: 0 m)	0.22+0.01	c	U15/888554	7	-67.0	151.5	1.9	1.1	338.1
NT99		Mamaku Ignimbrite	0.22+0.01	c	U16/739340	6	-65.1	147.8	2.0	0.0	334.8
	MP263	Mokai ignimbrite	0.22+0.01	AA[1]	T17/676006						
NT91		Mokai ignimbrite	0.22+0.01	c	T17/638035	4	-70.1	153.4	5.5	5.4	340.6
		Mean of Mamaku				6	-68.3	149.4	6.2	4.2	337.9
NT09		Rotorua lava dome			U16/945314	6	-49.9	12.2	4.7	77.5	233.2
NT10		Rotorua lava dome			U16/974329	6	-44.6	359.0	8.4	78.1	171.9
NT12		Rotorua lava dome			U15/996465	6	-60.0	55.9	6.0	47.5	288.3
NT55		Rotorua lava dome			U16/916402	6	-49.4	53.2	4.4	45.9	272.6
Maroa Centre											
NT44	MP43	Maroa lava dome	0.077+0.009	KA[5]	U17/728058	7	-54.8	7.3	5.9	83.4	240.3
NT34	MP28	Ongaroto Basalt	0.086+0.020	KA[5]	T17/667074	8	-82.1	23.0	4.2	52.3	346.1
NT32	MP72	Maroa lava dome	0.160+0.008	KA[5]	U16/732113	8	-55.7	4.5	4.6	85.8	237.2
NT33		Korotai Ignimbrite	< 0.22+0.01	sp	U17/702093	6	-61.3	341.8	4.8	75.6	64.1
NT36		Maroa lava dome	< 0.22+0.01	sp	U17/776005	5	-48.0	9.3	3.9	77.8	218.4
NT37		*Maroa lava dome*	< 0.22+0.01	sp	U17/817943	8	-58.4	27.7	5.2	68.5	276.3
NT08	MP39	Kakuki Basalt	0.221+0.036	KA[5]	U17/835957	5	-52.3	354.6	3.9	82.9	136.7
NT35		Atiamuri Ignimbrite			U17/763025	5	-56.4	1.6	3.7	88.1	217.6
NT14		Atiamuri Ignimbrite			U17/773043	6	-53.0	352.1	2.0	82.0	121.0
		Mean of Atiamuri				2	-55.0	356.6	13.7	86.1	131.7
NT97		Ohakuri Ignimbrite			U17/798059	-	-	-	>50	-	-
NT15		Maroa ignimbrite			U17/79023	6	-57.4	357.5	4.3	88.0	98.3
NT07		Tatua Basalt			U17/795952	7	-67.4	16.4	3.4	73.5	316.6
Reporoa Centre											
NT49	MP156	Kaingaroa Ignimbrite I	0.23+0.01	AA[1]	U16/086108	6	-47.0	6.3	1.0	78.6	205.7
NT17		Kaingaroa Ignimbrite II	0.23+0.01	c	V17/193054	6	-52.7	17.9	2.3	74.6	252.4
NT16		unidentified ignimbrite			U17/866025	6	-74.3	39.1	5.2	57.1	321.6
Okataina Centre											
NK12	MP157	Matahina Ignimbrite	0.28+0.01	AA[1]	V16/368146	9	-53.3	16.3	2.1	76.1	253.0
NT47		Okataina lava dome			U16/032237	6	-61.7	355.6	3.1	84.3	31.0
Whakamaru Centre											
NT46	MP14	Western Dome Belt	0.096+0.048	KA[5]	T17/548098	5	-45.6	349.4	6.8	75.6	134.5
NT13	MP04	Western Dome Belt	0.141+0.005	KA[2]	T17/639082	4	-53.6	34.8	7.4	61.8	268.1
NT45		Western Dome Belt			T17/598049	6	-65.9	13.0	3.0	76.4	316.2
NT06	MP37	Whakapapataringa dome	0.194+0.005	KA[5]	U17/853934	8	-71.1	34.8	5.3	61.3	314.0
	MP185	Mananui Ignimbrite	0.32+0.03	AA[1]	T17/300960						
NT53		Mananui Ignimbrite	0.32+0.03	c	T16/488282	6	-67.8	4.2	3.6	77.1	343.9
NT69		Mananui Ignimbrite	0.32+0.03	c	T17/334000	6	-61.5	11.5	4.1	80.3	294.9
		Mean of Mananui				2	-64.7	8.3	15.4	79.7	323.0
NT70	MP161	Paeroa Ignimbrite	0.33+0.01	AA[1]	U16/962102	6	-61.9	26.5	3.7	69.5	288.0
NT76		Te Whaiiti Ignimbrite	0.34+0.01	AA[1]	V19/150442	6	-63.4	357.2	1.9	83.6	14.7
NT05	MP183	Whakamaru Ignimbrite	0.32+0.02	AA[1]	T16/459136	8	-64.1	6.3	2.6	81.2	325.5
NT59		Rangitaikei Ignimbrite	0.32+0.02	c	T19/638404	6	-70.3	10.2	2.7	73.1	335.1
NT60		Rangitaikei Ignimbrite	0.32+0.02	c	U18/750549	8	-69.1	20.8	6.3	70.1	316.7
		Mean of Rangitaikei				2	-69.8	15.7	8.4	71.8	325.2

NT61		pre-Rangitaikei Ign	0.32+0.02	sp	U18/782525	6	-65.1	355.7	2.5	81.2	15.4
NT78		Aratiatia lava dome	> 0.33+0.01	sp	U17/836827	7	-70.1	1.9	12.5	74.4	352.0
NT71	MP258	Te Kopia Ignimbrite	0.34+0.01	AA[1]	U17/951096	7	-63.8	38.8	1.9	60.6	292.7
NT11	MP01	Western Dome Belt	0.397+0.016	KA[2]	T17/656087	6	-38.2	339.6	6.7	65.6	123.9
Kapenga Centre											
NT48		Earthquake Flat Breccia			U16/017226	6	-62.2	8.5	9.7	81.7	308.3
NT04		Pokai Ignimbrite			T16/683268	7	-57.2	354.6	7.5	85.7	89.5
NT98		Chimpanzee ignimbrite			T16/658332	6	-52.6	6.4	1.7	82.8	223.8
NT01		Waiotapu Ignimbrite	0.71+0.06	c	T16/627174	6	-56.1	348.8	4.3	81.0	93.1
NT21	MP26	Waiotapu Ignimbrite	0.71+0.06	AA[1]	U16/770212	7	-62.4	348.8	3.1	80.0	49.6
NT92		Waiotapu Ignimbrite	0.71+0.06	c	T16/588400	6	-62.9	2.4	3.1	83.5	341.7
NT03		Waiotapu Ignimbrite	0.71+0.06	c	U17/018098	6	-64.5	24.2	2.4	70.5	298.4
		Mean of Waiotapu				4	-62.2	359.9	9.8	84.6	353.7
NT24		? Waiotapu Ig.	0.71+0.06	pmc	U16/788203	6	-58.7	356.5	2.1	87.1	61.8
NT27		Haparangi lava dome			U16/906246	8	-82.0	333.2	2.5	51.8	7.7
NT101		Rahopeka Ignimbrite	0.77+0.03	c	U16/754185	4	59.4	159.5	4.7	-74.0	252.8
NT102		Rahopeka Ignimbrite	0.77+0.03	c	U16/812186	4	51.5	144.6	7.7	-60.6	267.3
NT105	MP169	Rahopeka Ignimbrite	0.77+0.03	AA[1]	U16/786201	4	46.9	171.7	5.2	-77.7	319.4
		Mean of Rahopeka				3	52.7	166.9	11.1	-78.3	287.4
NT26		Pukerimu Ignimbrite	0.89+0.04	c	U16/810215	8	32.1	184.6	5.5	-68.8	8.3
NT25	MP167	Tikorangi Ignimbrite	0.89+0.04	AA[1]	U16/792206	9	38.0	177.8	2.4	-72.9	349.0
Mangakino Centre											
NT54		Marshall-A (Col: 40m)	0.95+0.03	c	T16/541234	8	65.9	171.0	3.0	-78.1	206.2
NT87		Marshall-B (Col: 30m)	0.95+0.03	c	T16/541234	5	60.0	189.7	2.4	-82.1	108.4
NT86		Marshall-B (Col: 25m)	0.95+0.03	c	T16/54`134	6	70.7	183.1	2.4	-73.1	169.6
NT85		Marshall-B (Col: 0m)	0.95+0.03	c	T16/541234	5	65.7	170.2	6.5	-78.0	209.2
NT84		Marshall-A (Qua: 20m)	0.95+0.03	c	T16/579203	6	62.0	185.2	1.8	-83.7	139.1
NT79		Marshall-A (Qua: 15m)	0.95+0.03	c	T16/579203	5	67.8	180.3	2.7	-77.6	174.8
NT80		Marshall-A (Qua: 12m)	0.95+0.03	c	T16/579203	5	62.8	169.6	5.3	-80.2	225.2
NT81		Marshall-A (Qua: 4m)	0.95+0.03	c	T16/579203	5	62.3	177.0	3.1	-84.2	198.0
NT82		Marshall-A (Qua: 1m)	0.95+0.03	c	T16/579203	6	63.4	178.4	1.1	-83.2	185.6
NT83		Marshall-B (Qua: 0m)	0.95+0.03	c	T16/579203	6	58.1	174.7	3.0	-85.8	256.8
NT02		Marshall-B Ignimbrite	0.95+0.03	c	T16/594186	6	57.7	215.3	2.9	-62.4	97.1
NT51	MP174	Marshall-B Ignimbrite	0.95+0.03	AA[1]	T16/512258	6	61.5	189.0	2.6	-81.9	121.3
		Mean of Marshall				12	63.7	182.5	3.8	-82.5	163.8
NT88		Kaahu ignimbrite			T17/560039	6	53.6	186.8	4.8	-83.1	49.8
NT40	MP190	Rocky Hill Ignimbrite	1.00+0.05	AA[1]	S17/185955	6	-51.2	8.8	2.6	80.2	225.2
NT90	MP90	Whaakahu lava domeS	1.020+0.017	KA[5]	T17/579967	5	81.1	44.7	14.3	-25.3	189.4
NT89		Whakaahu lava domeN			T17/566028	5	67.4	164.5	5.2	-73.9	214.0
NT50	MP164	Mangakino Lava Dome	1.27+0.05	AA[4]	T16/522213	7	67.9	166.3	3.0	-74.1	209.1
NT31		Ahuroa Ignimbrite	1.18+0.02	c	T17/402077	9	78.4	8.7	5.0	-16.3	179.1
NT39		Ahuroa Ignimbrite	1.18+0.02	c	T18/393075	8	66.6	347.3	5.4	1.7	167.3
NT63	MP189	Ahuroa (Wai: 20m)	1.18+0.02	AA[1]	S17/190021	6	81.3	235.0	8.1	-46.6	155.0
NT74	MP189	Ahuroa (Wai: 17m)	1.18+0.02	AA[1]	S17/190021	5	85.5	54.1	2.7	-32.8	184.1
NT42	MP189	Ahuroa (Wai: 14m)	1.18+0.02	AA[1]	S17/190021	9	85.8	125.5	3.9	-43.0	184.7
NT64	MP189	Ahuroa (Wai: 10m)	1.18+0.02	AA[1]	S17/190021	6	67.5	22.5	14.2	-1.1	189.5
NT75	MP189	Ahuroa (Wai: 2m)	1.18+0.02	AA[1]	S17/190021	5	64.1	351.4	21.1	5.4	169.4
NT65	MP189	Ahuroa (Wai: 0m)	1.18+0.02	AA[1]	S17/190021	-	-	-	-	-	-
NT52		Ahuroa Ignimbrite	1.18+0.02	c	T16/490288	10	88.7	44.6	4.0	-36.4	177.9
NT68		Ahuroa (Gul: 10m)	1.18+0.02	c	T17/333998	5	81.1	62.3	3.5	-29.1	193.1
NT67		Ahuroa (Gul: 3m)	1.18+0.02	c	T17333998	7	78.0	41.0	8.3	-19.9	191.5
		Mean of Ahuroa				10	80.4	10.4	7.5	-21.3	179.3
NT41		Ongatiti Ignimbrite	1.21+0.04	c	S17/145015	6	66.2	183.0	4.5	-79.7	164.1
NT66	MP187	Ongatiti Ignimbrite	1.21+0.04	AA[1]	S17.143015	6	72.0	174.6	2.8	-71.1	184.5
		Mean of Ongatiti				2	69.2	179.4	14.2	-75.6	177.3
NT73		Waipari Ignimbrite			T16/354336	8	67.3	175.5	1.8	-77.6	189.1
NT72	MP197	Wharepuhunga ignimbrite	1.44+0.03	AA[4]	S16/267324	6	69.6	185.8	5.8	-74.3	162.7
NT43		Ignimbrite F	1.60+0.09	KA[3]	T17/425910	7	64.0	172.3	2.8	-80.8	211.4
NT38	MP191	Ignimbrite C	1.68+0.07	AA[1]	T18/387079	7	57.3	185.8	5.4	-85.4	81.5
NT29	MP192	Ignimbrite B	1.53+0.04	AA[1]	T16/355166	6	58.3	185.8	3.1	-85.4	95.8
	MP186	Ignimbrite A	1.55+0.05	AA[1]	S17/125922						
NT28		Ignimbrite A	1.55+0.05	c	T16/347193	8	61.7	189.7	2.6	-81.3	120.3
NT40		Ignimbrite A	1.55+0.05	c	T16/364173	8	61.9	188.9	2.2	-81.7	123.7
		Mean of Ignimbrite A				2	61.8	189.3	0.9	-81.5	122.0

* Rolles Peak is a small independent cone: it predates, and is not associated with the activity of any of the nearby major volcanic centres of TVZ.

AA[1] Ar/Ar age from Houghton et al., 1995
KA[2] K/A age from Houghton et al., 1991
KA[3] K/A age from Soengkono et al., 1992
AA[4] unpublished Ar/Ar ages by McWilliams & Houghton
KA[5] unpublished K/Ar ages by Lanphere & Houghton
c unambiguous correlation with directly dated unit
pmc correlation on basis of palaeomagnetic result
sp age estimate on basis of stratigraphic position